Resonant Gravity Triadic Mode

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PAPER_962: Resonant Gravity Triadic Mode

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 215 **Source:** triadic_solutions_next.py (ResonantGravityTriadic) **Calculator:** ResonantGravityTriadicCalc (CP4 #546) **CVW:** v2.0.0 compliant

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Abstract

The Resonant Gravity Triadic mode uses the 1.25 THz phonon linewidth Γ to tune neutron-drop and buoyancy reversal. When $\Phi(\omega_t extSCm, \Gamma) > \Phi_t extcrit$, the neutron drip-line shifts, controlling NS merger dynamics.

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1. Resonant Phonon Occupation

$$\Phi(\omega, \Gamma) = \Phi_0 \cdot \text{exp!} \left(-\frac{(\omega - \omega_t extSCm)^2}{2\Gamma^2} \right) \cdot S_{26}([SSq])$$

2. Neutron-Drop Threshold

$$\Phi(\omega_t extSCm, \Gamma) > \Phi_t extcrit \implies \text{neutron-drop triggered}$$

3. Buoyancy Reversal

$$t_{\text{rev}} = \frac{\pi}{2\Gamma}$$

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_961 — Compressed Gravity Triadic
3. PAPER_963 — Buoyancy Gravity Triadic
4. PAPER_966 — Unified Triadic Solver

Cross-Links

Related Paper	Relationship
PAPER_961	Compressed branch of same triadic
PAPER_963	Buoyancy branch
PAPER_966	Unified solver combining all three
PAPER_955	Phonon resonance frequency link

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	5.0×10^{-4} day^{-1}	Damping
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy
$\Phi_t extcrit$	—	Neutron-drop threshold	Phase transition

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
g_{res} peak at $\omega_t extSCm$	Resonance amplification	Derived
$\Phi_t extcrit$ neutron-drop	Phase boundary for neutron star cores	Novel

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Resonant Gravity (5-Frequency Multi-Mode)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{res}} = g_{\text{comp}}(r) \cdot \prod_{f \in \{\text{Super,Quantum,Aether,Fluid,Exp}\}} \big(1 + A_f \sin(\omega_f t + \phi_f)\big)$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{g_{\text{res}}(r,t) = g_{\text{comp}}(r) \prod_{f=1}^{5} (1 + A_f \sin(\omega_f t + \phi_f)), \quad \Phi_t extcrit : g_{\text{res}} > g_{\text{neutron-drop}}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 → compressed gravity → 5-frequency modulation → resonant amplification → $\Phi_t ext crit$ phase boundary

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§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

Resonance peaks modulate VDS by factor $\prod(1 + A_f)$.

§B.2 DVP

Five frequencies correspond to five dipole vortex modes.

§B.3 BSH

g_{res} bounded above by $g_{\text{comp}} \cdot \prod(1 + A_f)$ (constructive limit).

§B.4 Production-Scale Consistency

Metric	Value	Status
5 frequencies	All modulated	Confirmed
$[SSq]$	0.57	Confirmed